

PARTICLE ZOO

While electrons are truly elementary particles, which cannot be divided any further (and part of a family of particles called the leptons), protons and neutrons are made up of three even smaller particles called quarks (see p.122). Particles formed by groups of quarks are collectively known as hadrons, which are subdivided into baryons (made up of triplets of quarks) and mesons (made up of a paired quark and antiquark particle).

FERMIONS

The elementary (indivisible) particles that make up matter fall into two groups: leptons and quarks. Only a few of each family are widespread in today's universe.

The subatomic world

Using particle accelerators (see p.121) to break apart atoms and create short-lived and unstable particles, physicists have assembled the so-called Standard Model (see pp.124–125) of particle physics.

